

LUBNY, Ukraine

WMO: 333770

Lat: 50.0158N

Lon: 32.9867E

Elev: 518

StdP: 14.42

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	4.8	-6.7	3.9	1.4	-1.7	5.1	5.8	16.7	24.8	14.4	24.9	4.0	0	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	88.9	68.9	85.8	67.5	82.8	66.3	71.1	83.5	69.5	81.6	68.2	79.7	4.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.0	101.1	77.0	65.4	95.6	75.0	64.0	91.1	73.6	35.3	83.7	33.9	81.6	32.8	80.0	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.6	11.2	9.6	-6.0	92.9	6.1	3.5	-10.4	95.4	-13.9	97.5	-17.4	99.4	-21.8	102.0		
(5)			-6.5	73.6	6.0	1.9	-10.8	75.0	-14.3	76.1	-17.6	77.2	-22.0	78.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	23.8	26.0	35.1	49.8	60.5	66.9	70.4	68.6	58.7	46.7	36.3	27.6	(6)
(7)		DBStd	18.53	9.84	10.11	8.03	7.73	8.05	6.56	5.55	6.53	7.37	7.96	8.41	9.53	(7)
(8)		HDD50	3333	813	671	462	96	12	0	0	0	12	158	415	694	(8)
(9)		HDD65	6843	1278	1091	925	457	185	57	15	34	215	567	860	1159	(9)
(10)		CDD50	2479	0	0	2	90	338	507	631	578	272	57	4	0	(10)
(11)		CDD65	517	0	0	0	2	47	114	181	147	25	1	0	0	(11)
(12)		CDH74	4430	0	0	0	19	425	925	1479	1360	217	5	0	0	(12)
(13)		CDH80	1323	0	0	0	3	93	236	462	484	45	0	0	0	(13)
(14)	Wind	WSAvg	4.4	5.0	5.1	5.2	4.7	4.1	4.0	3.7	3.6	3.7	4.0	4.9	5.0	(14)
(15)	Precipitation	PrecAvg	24.6	2.0	1.7	1.7	1.7	1.7	2.7	2.8	2.6	1.8	1.5	2.2	2.3	(15)
(16)		PrecMax	38.3	5.6	3.7	4.0	4.0	4.5	6.6	6.4	6.0	6.9	5.3	4.6	4.1	(16)
(17)		PrecMin	16.9	0.2	0.1	0.1	0.4	0.3	0.5	0.3	0.4	0.2	0.0	0.3	0.2	(17)
(18)		PrecStd	5.1	1.2	0.9	1.0	0.9	0.9	1.6	1.5	1.3	1.6	1.3	1.0	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	47.4	60.7	76.3	87.0	89.0	91.9	95.4	84.9	72.8	59.5	49.0	(19)
MCWB			41.2	42.4	47.7	57.5	65.0	67.7	70.2	68.0	64.2	59.0	54.3	47.6	(20)	
2%		DB	40.1	42.6	54.9	70.7	82.4	85.6	88.5	89.6	80.2	67.6	53.7	44.7	(21)	
		MCWB	38.3	40.1	45.4	54.5	62.8	67.1	69.7	69.2	62.3	57.4	50.5	42.9	(22)	
5%		DB	37.2	39.7	50.3	66.6	79.1	82.7	85.3	85.3	75.3	62.2	50.1	41.6	(23)	
		MCWB	35.8	37.8	43.2	53.0	62.2	66.3	68.4	67.6	60.7	55.4	48.0	40.4	(24)	
10%		DB	35.2	37.2	46.2	62.9	74.9	79.7	82.0	81.4	71.3	58.0	47.3	38.6	(25)	
		MCWB	34.2	35.4	41.0	51.1	60.6	65.3	67.6	65.6	59.1	53.0	45.5	37.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	43.7	50.0	59.6	68.0	71.4	73.4	73.3	67.1	62.7	54.4	47.8	(27)
MCD B			43.1	46.1	58.2	73.8	80.9	82.6	85.8	86.2	80.2	68.0	58.4	48.8	(28)	
2%		WB	38.4	40.7	46.6	56.4	65.5	69.4	71.7	70.5	64.3	58.8	51.1	43.3	(29)	
		MCD B	39.6	42.3	52.8	66.9	78.0	81.2	84.3	83.9	74.8	64.5	53.2	44.2	(30)	
5%		WB	36.0	37.9	43.9	54.4	63.7	68.0	70.1	68.7	62.3	56.1	48.3	40.5	(31)	
		MCD B	36.9	39.3	49.0	63.4	75.7	79.5	81.8	82.1	72.2	61.2	49.7	41.6	(32)	
10%		WB	34.4	35.7	41.4	52.6	61.8	66.4	68.7	67.2	60.3	53.4	45.7	37.7	(33)	
		MCD B	35.2	36.8	45.8	60.9	72.5	77.2	79.3	79.1	68.9	57.4	47.2	38.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	9.3	11.6	16.1	18.4	17.1	17.2	18.8	16.6	13.5	8.2	7.4	(35)
MCDBR			7.3	10.1	18.4	21.2	22.6	21.0	21.1	23.4	23.1	20.4	11.7	8.7	(36)	
MCWBR			6.5	8.1	11.0	10.0	8.9	8.4	7.9	8.4	10.0	11.4	9.1	7.9	(37)	
5% WB		MCD BR	6.6	9.4	16.4	18.0	20.5	18.6	18.6	20.5	19.9	17.2	10.4	8.3	(38)	
		MCWBR	6.2	8.1	10.8	9.3	9.1	8.5	7.9	8.7	10.2	11.2	8.9	7.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.314	0.344	0.376	0.441	0.408	0.400	0.429	0.428	0.399	0.363	0.346	0.317	(40)	
taud		2.185	2.126	2.172	2.099	2.276	2.314	2.241	2.230	2.278	2.354	2.299	2.258	(41)		
Ebn at Noon		221	244	259	253	270	273	262	255	250	238	209	203	(42)		
Edh at Noon		24	33	38	46	40	39	42	40	34	27	22	20	(43)		
(44)	All-Sky Solar Radiation	RadAvg	297	558	942	1399	1837	1957	1905	1685	1126	648	265	195	(44)	
(45)		RadStd	56	85	152	123	158	170	108	113	155	96	45	47	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.37	N/A	N/A	+1.91	N/A	N/A	N/A	-378	+261	+164
(47) Regional (11 neighbors)		+1.31	N/A	N/A	+1.78	N/A	N/A	-252	-353	+209	+113

Nomenclature: See separate page